

Kieron von Buchstab

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Skills

CAD: CATIA V5, SolidWorks, Fusion 360, OnShape, KiCAD

Analysis: Ansys WorkBench, BladeGen, Ansys Electronic Desktop

Experience

Systems and Software Engineering Intern, Lockheed Martin - Ottawa, ON May 2023 – Sept 2025

- Optimized certification workflow by using Python and Pandas to analyze large regulatory datasets to extract relevant information for each sub-team
- Directed laser systems certification effort, coordinating across Lockheed Martin, Irving Shipbuilding, and National Defence to ensure all lasers onboard the Canadian River Class Destroyer were certified for use
- Developed Python script to automate parsing of rules and regulations kept in WindChill (document management client) saving estimated 500 hours of manual labor
- Developed Python/VBA tool for turning Excel sheets into Visio diagrams to display regulatory compliances at vendor meetings, ultimately saving 4h of work every week

CubeSat Research Assistant, Carleton University – Ottawa, ON Nov 2021 – Present

- Designed and manufactured a mobile ground station for tracking fast moving satellites in Low Earth Orbit (LEO) using SolidWorks, C++, and Python, leading to the successful receipt and decoding of weather data from LEO satellites as well as a geostationary satellite
- Developed ground station active tracking software in Python using peak signal power analysis to search for difficult to track satellites allowing for the successful discovery and reception of geostationary satellites
- Constructed and validated dipole and yagi antennas for communication with satellites in Low Earth Orbit, culminating in the successful reception and decoding of weather data from NOAA satellites

Projects

5.8GHz FMCW Short-range Radar

- Designed and fabricated a 5.8 GHz Frequency-Modulated Continuous Wave (FMCW) RADAR for Martian subsurface mapping; Aim to achieve 1-meter range resolution for autonomous cave exploration and geological surveying
- Validated high-frequency PCB performance using Ansys Electronics Desktop; Achieved optimized impedance matching and verified active component functionality prior to fabrication to eliminate design iterations
- Currently developing C++ algorithms to process FMCW chirp frequencies; implementing Fast Fourier Transforms (FFT) to achieve a 1-meter range resolution for real-time target mapping

Liquid Fuel Jet Engine

- Fabricated a self-sustaining turbojet engine utilizing an automotive turbocharger, generating 100 Newtons of thrust and demonstrating practical combustion and fluid dynamics principles outside of a classroom setting
- Modeled and manufactured a custom flame tube and combustion chamber from stainless steel using CATIA, calculating air intake and cooling hole patterns to ensure the required air-fuel ratio for sustained combustion
- Conducted a mass flow and thermodynamic analysis on the system to determine a precise fuel flow rate, successfully integrating the fueling and oiling system to sustain the engine's target operational air-fuel ratio

Publications

von Buchstab, K., Jurgutis, A., Lear, A., Jazebizadeh, H., & Burlton, B. (2022, September 2). *The Receipt and Analysis of Weather Data in a Simulated Martian Environment*. International Astronautical Congress 2022.

Jurgutis, A., Lear, A., Murray, M., **von Buchstab, K.**, Jazebizadeh, H., & Burlton, B. (2022, September 20). *Design, Verification, and Validation of the Communication System of an Undergraduate CubeSat Mission*. International Astronautical Congress.

Education

Carleton University – BEng in Aerospace Engineering

April 2026